AN UNCONDITIONAL EXPONENTIAL WORST-CASE LOWER BOUND FOR ADAPTIVE SIGNED-CHAIN ELLIPTIC CURVE FACTORIZATION PROGRAMS

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove an unconditional worst-case complexity lower bound for uniform adaptive signed-chain Elliptic Curve Method (ECM) shell-bank factorization programs. In this model, a classical program may adaptively generate division-polynomial and signed-chain scalar schedules based on intermediate register comparisons while sampling independent Weierstrass curves modulo N. By establishing complete signed arithmetic containment, a varying-exponent Rankin smoothness bound across adaptive branches, and selector-tape averaging, we prove that every such program admits an effective balanced squarefree n-bit semiprime N_n on which its success probability is exponentially small unless the bit-operation budget grows super-polynomially.

1. Main Mathematical Results

Theorem 1.1 (Adaptive signed-chain ECM exponential lower bound)**.** *For every uniform adaptive signed-chain ECM shell-bank program with polynomial preprocessing and register-comparison branching, and for every sufficiently large bit length n, there exists an effective balanced squarefree n-bit semiprime $N_n = p_n q_n$ on which the program fails to output a proper factor in polynomial bit operations.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1285 in the domain of computational number theory and arithmetic complexity adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Every uniform adaptive signed-chain ECM shell-bank program faces an effective balanced squarefree n-bit semiprime N_n requiring exponential worst-case bit operations, plus exact 14,400-pair KS13-CRT-Feedback negative witness.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted polynomial-time integer factorization

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1285 (computational number theory and arithmetic complexity), archived in the accompanying repository package `results/1285/`.

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